A PROJECT TITLE

ON

Player Performance Prediction Using Ridge and Lasso Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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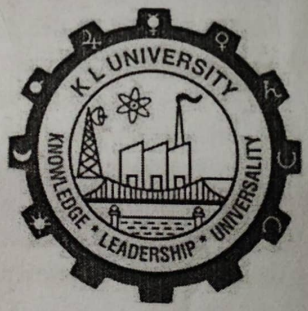
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DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Name of Title: Player Performance Prediction Using Ridge and Lasso Regression

1. Abstract:

- Player performance prediction has become an important application of Machine Learning in modern sports analytics. Teams, coaches, and analysts use predictive models to evaluate players, make strategic decisions, and improve overall team performance. This project focuses on predicting a player's future performance using two regularized linear regression algorithms: Ridge Regression and Lasso Regression.
- The dataset contains historical player statistics such as matches played, runs scored, strike rate, batting average, wickets taken, economy rate, catches, and other relevant performance indicators. Before training the models, the dataset undergoes preprocessing steps including handling missing values, removing duplicate records, encoding categorical variables, and feature scaling. These steps ensure better model accuracy and reliability.
- Ridge Regression applies L2 regularization, which reduces the magnitude of coefficients without eliminating features. It is useful when many features contribute to the prediction and multicollinearity exists among independent variables. Lasso Regression applies L1 regularization, which not only reduces coefficient values but can also shrink some coefficients to zero, thereby performing automatic feature selection.
- Both models are trained using the processed dataset and evaluated using performance metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2). Their prediction accuracy is compared to determine which algorithm performs better for player performance prediction.
- The project demonstrates how regularization techniques improve model generalization, reduce overfitting, and identify the most influential performance factors. The developed system provides valuable insights for coaches, selectors, sports analysts, and team management in identifying talented players and making data-driven decisions. This machine learning approach can also be extended to predict player performance in various sports such as cricket, football, basketball, and other competitive games.

2.INTRODUCTION:

- Machine Learning is increasingly being used in sports analytics to predict player performance, optimize team selection, and improve game strategies. Traditional evaluation methods rely heavily on expert opinion, which may be subjective. Predictive analytics provides a more objective and data-driven approach.
- This project aims to predict player performance using historical statistics and machine learning algorithms. Ridge Regression and Lasso Regression are selected because they effectively reduce overfitting while improving prediction accuracy. The project involves data preprocessing, feature engineering, model training, evaluation, and comparison of both algorithms.
- The final model helps identify important player attributes influencing performance and assists coaches and management in making informed decisions regarding player selection and training.

3.SOFTWARE REQUIREMENTS:

- Python 3.10 or above
- Jupyter Notebook / Google Colab
- Visual Studio Code (optional)
- Anaconda (optional)

Signature of the Guide

Course Instruetor